RIDHI SRIKANTH

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SUMMARY

UC San Diego Computer Engineering Honors student (3.99 GPA) focusing on programming languages and compiler research. Currently extending the CPython compiler and runtime to enable more efficient execution of system-level tasks through language-level abstractions. Aiming to leverage low-level systems expertise to build robust, reproducible computational infrastructure.

EDUCATION

University of California, San Diego | BS, Computer Engineering | **GPA:** 3.99 | **Graduation:** June 2027

- **CSE Dept. Honors Candidate:** Faculty-advised research track culminating in an Honors thesis.
- **Awards:** Eta Kappa Nu (HKN) IEEE Honor Society; Provost Honors (8 consecutive quarters), 1st Place, UCSD WIC Programming Competition (Fall 2023)
- **Selected Research Programs:** Brown Systems Week 2026; Cornell–Maryland–Max Planck Pre-doctoral Research School (CMMRS) 2026
- **Selected Coursework:**DSA, Operating Systems, Parallel Programming, Computer Architecture, Machine learning, Networks

RESEARCH EXPERIENCE

Research Lead - UC San Diego CSE Department - (Advisor: Professor Michael Coblenz)
Enhancing Shell Scripting Workflows by Modifying Python Compiler in C June 2025 - Present

- Leading an independent research project to improve the safety and usability of shell scripting for Python programmers
- Performed thematic and static analysis of shell scripts to inform the design of Python language extensions
- Extending the CPython compiler pipeline across PEG parsing, AST construction, bytecode generation, and evaluation
- Implementing runtime semantics via C-level extensions to the CPython evaluation loop to support new language constructs
- Debugging low-level compiler/runtime issues using GDB and backward tracing
- Developing a shell-to-extended-Python translation system to evaluate expressiveness using the Koala benchmark against native shell scripts; Designing usability and effectiveness user studies for the extended language features
- Contributing to a research manuscript (targeting publication)

Research Assistant - SALAD Lab UCSD
Scientific Workflow Reproducibility Study Feb 2025 - July 2025

- Collaborated with a research team on a corpus study of 500+ scientific programs
- Analyzed selected programs for runtime errors, dependency issues, and execution failures affecting software reproducibility
- Contributed to a research manuscript on scientific workflow reproducibility (unpublished)

TECHNICAL EXPERIENCE

ECE 35 (Analog Design) Course Tutor Sep 2025 - April 2026

- Managed support for 400+ students, providing tutoring and hands-on lab guidance on real device circuit implementation
- Collaborated in weekly meetings to refine instructional strategies and was responsible for grading quizzes and final exam questions on fundamental circuit theory concepts.

Full-stack Developer - ElectroTriton With Professor Curt Schurgers | [ElectroTriton](#) Aug 2025 - Present

- Developing features in ElectroTriton, a Django and MySQL homework management web app used for Engineering courses
- Implemented course cloning and automated roster updates, reducing management time by ~71% and ~78%, respectively
- Developed a new analytics dashboard for instructors to track assignment metrics, including view, start, and completion rates.

Full-stack Developer - LabStream With Professor Saharnaz Baghdadchi | [LabStream](#) Oct 2024 - June 2025

- Engineered a React/Firebase web app with a team of 8 to enable real-time remote access to electrical engineering labs
- Architected a slot-booking system reducing lab scheduling time by 65%, role-based access control, user session management

SKILLS

- **Programming Languages**: OpenCL, Python, C, C++, TypeScript, JavaScript, HTML, SQL, CSS, Shell Scripting, Rust, Java
- **Technologies and Expertise**: React, Flask, Django, Agile, CI/CD, Git, Linux, Embedded Systems, Firebase

LEADERSHIP

- IEEE UCSD Chapter Vice Chair Events (2026–27): Lead the events team to organize technical, professional, outreach, and social events for the UC San Diego IEEE community, partnering with industry and campus organizations.
- IEEE UCSD Chapter Technical Chair (2024-26): Led 5+ technical workshops for 100+ students, including live-coding sessions on React and recommendation systems, fostering hands-on learning for peers

PROJECTS

Hardware Developer | [Hardware Blog](#): Designed and prototyped 5+ embedded systems with C++ firmware and custom PCBs.